EMNLP 2026 · INDUSTRY TRACK · PAPER 194

SWE-Tester

Training open-source LLMs for issue reproduction in real-world repositories

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MOTIVATION

Bug reports rarely ship with a test that reproduces them

Developers

A failing test pins down the bug and speeds up root-cause analysis. Writing one by hand is tedious.

Coding agents

Reproduction tests enable test-first workflows and let issue-resolution agents check their own patches.

The gap

Strong methods rely on closed APIs (GPT-4o, Claude). Sending proprietary code to a third party raises IP and privacy concerns.

TASK DEFINITION

Write tests that fail before the fix and pass after it

Input: repository R at the pre-PR commit, plus the natural-language issue I

Output: a test set T, written as an edit to an existing test file

Judged with the hidden gold fix X*, which the model never sees

Unlike issue resolution: prior work finds no correlation between fixing a bug and reproducing it

T REPRODUCES THE ISSUE IF

≥ 1 test is F→P

fails on the buggy code, passes with the fix

no F→F, no P→F

P→P regression tests are allowed

COMPARISON ON SWT-BENCH VERIFIED

A fully open 32B model, competitive with GPT-4o pipelines

Method	Model	Success (%)
ZeroShotPlus	GPT-4o	14.30
LIBRO	GPT-4o	17.80
SWE-Tester 7B	Qwen2.5-7B	8.31
SWE-Tester 14B	Qwen2.5-14B	11.79
SWE-Tester 32B	Qwen2.5-32B	14.09

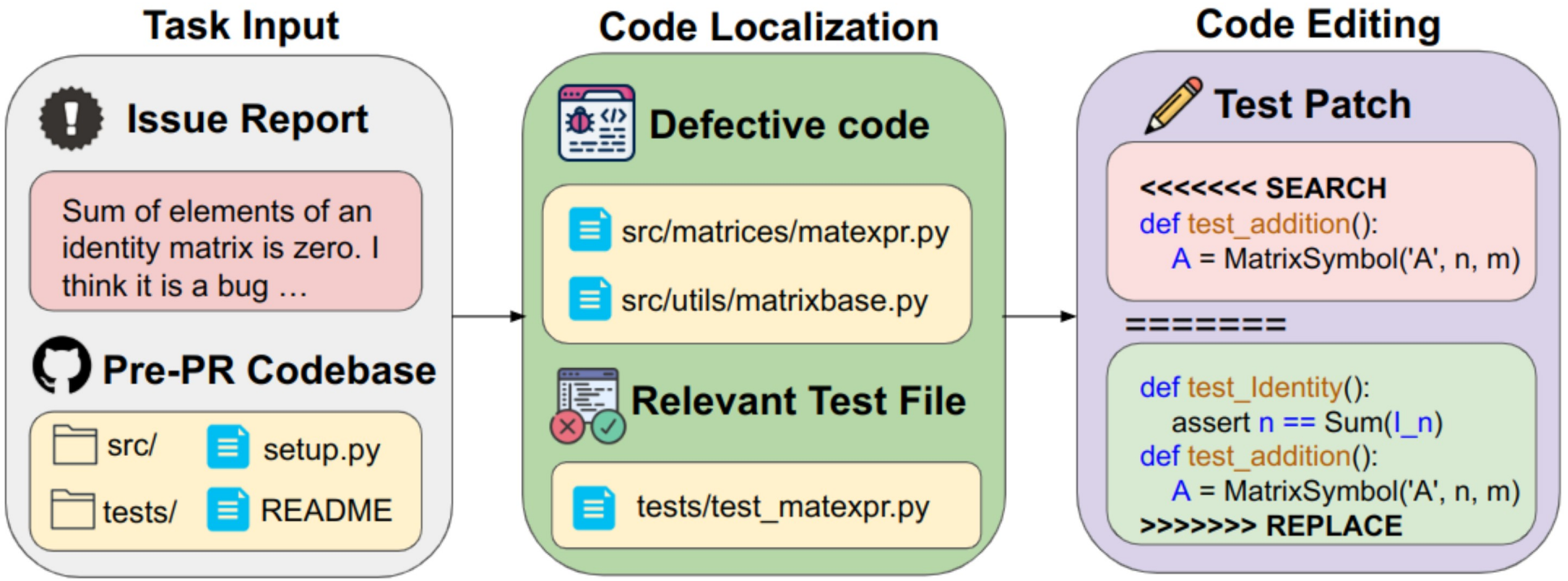
Why it matters in industry

Open weights you host: code never leaves your network

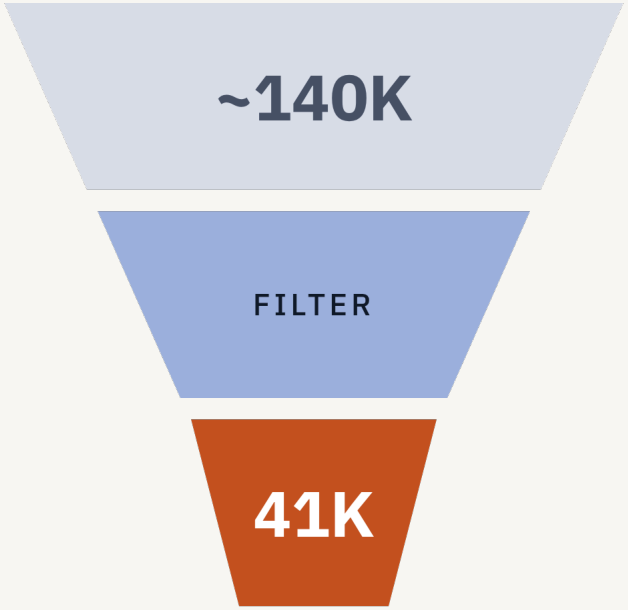
Simple, non-agentic pipeline: predictable cost and latency

APPROACH

Training Data Curation Pipeline



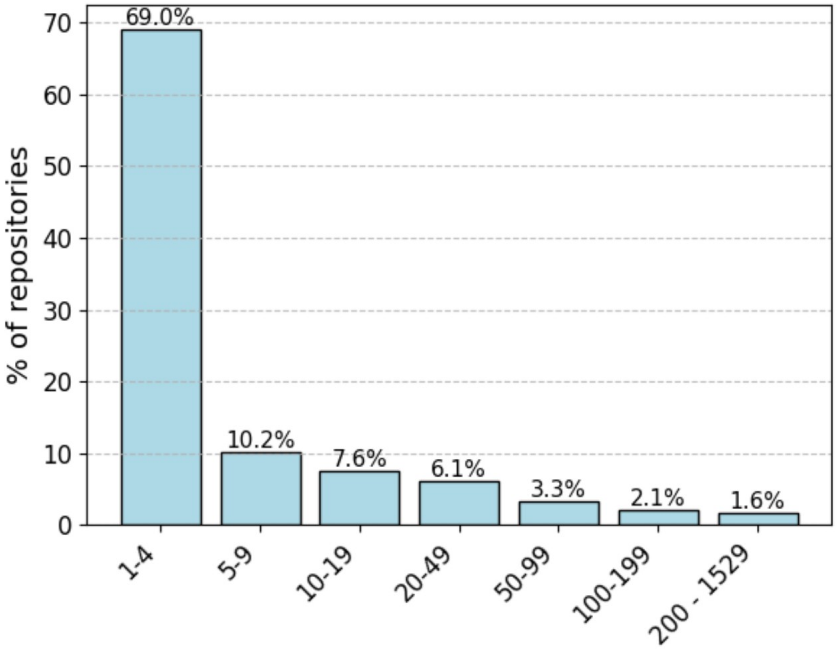
Training Data Statistics



Raw issue–PR pairs
from ~5K repositories

Python only
1–3 source files, exactly 1 test file
SWT-Bench repos removed
De-duplicated

Training instances
from 2.6K repositories · ~29% kept



Target: the PR's own test-file edit, as Search/Replace

CONTROLLED REARING

Plain supervised fine-tuning, five models, three families

Models

- Qwen2.5-Coder-Instruct 7B / 14B / 32B
- Llama-3.1-8B-Instruct
- Gemma-3-12B-Instruct

Recipe

- Prompt: issue + gold source files + test file
- ≤ 16K tokens → 23K instances used
- 2 epochs, batch 32, AdamW, cosine LR
- LR 3e-5 (1e-5 for Gemma)
- 2–8 A100s
- SWT-Bench Verified as Evaluation Dataset

No distillation from a closed teacher, and no RL: the signal is developer-written tests.

SWT-BENCH VERIFIED (433 ISSUES)

Inference Scaffold for Evaluation



Source

Localize defective code

SWE-Fixer 7B Retriever retrieves the defective source files

Xie et al., 2025

Top-4

Rerank test files

Qwen3-Embedding-8B ranks candidate test files; keep the top K = 4, since the top-ranked one may be wrong

Zhang et al., 2025

32

Sample candidates

N = 8 patches at T = 0.5 for each of the 4 test files, all sharing the same SWE-Fixer source context

Only the test file varies: test localization matters more than source localization (Mündler et al., 2024; Ahmed et al., 2025a).

RESULTS

Fine-tuning lifts every model on every metric

Model (Base / SFT)	Applicability 0–100	Change coverage 0–25	Success rate 0–15
Llama-3.1-8B	18.0 → **84.8**	1.4 → **22.3**	0.2 → **6.2**
Gemma-3-12B	85.9 → **91.0**	6.0 → **24.1**	0.9 → **8.1**
Qwen2.5-Coder-7B	40.4 → **85.0**	3.2 → **21.3**	1.2 → **8.3**
Qwen2.5-Coder-14B	31.0 → **85.0**	2.5 → **23.5 (+21.0)**	1.4 → **11.8 (+10.4)**
Qwen2.5-Coder-32B	84.8 → **84.8**	19.1 → **22.7**	9.9 → **14.1**

91% / 8%

Gemma-3-12B: 91% of patches apply, only 8% reproduce the bug. Valid edits are not valid tests.

≤ 1.4%

Base success for every model at 14B or smaller. Off-the-shelf small models can't do this task.

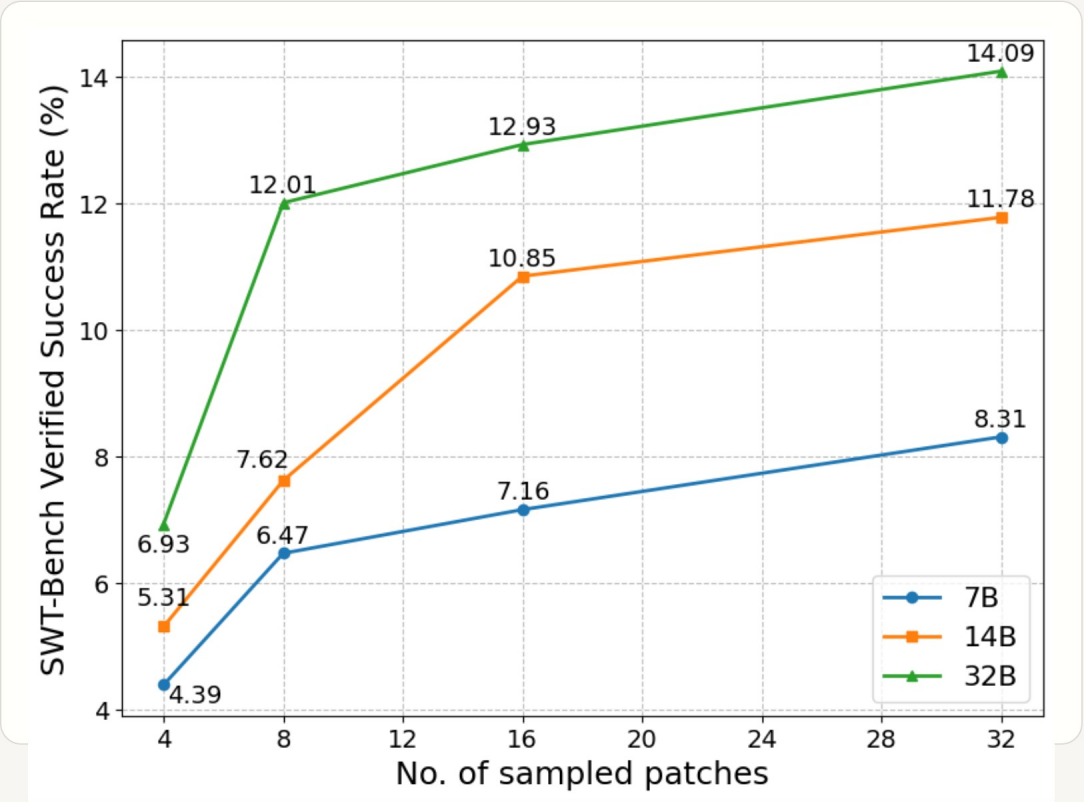
+21 pts

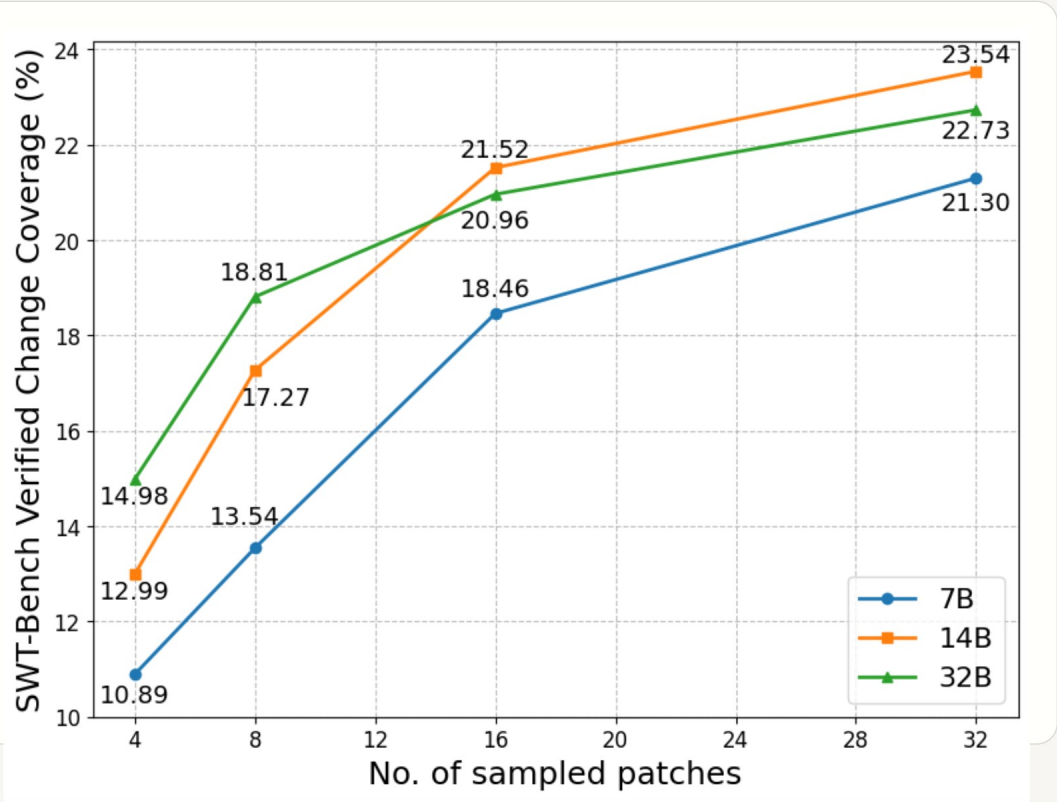
Change coverage for Qwen2.5-Coder-14B: tests now exercise the code the fix touches.

All values %, same inference pipeline; bars use each column's header scale. Applicability = patch applies cleanly; change coverage = share of gold-fix lines the new tests execute; success = issue reproduced (F→P, nothing broken).

NO SATURATION UNTIL 32 PATCH SAMPLES: POSSIBLE IMPROVEMENT HEADROOM

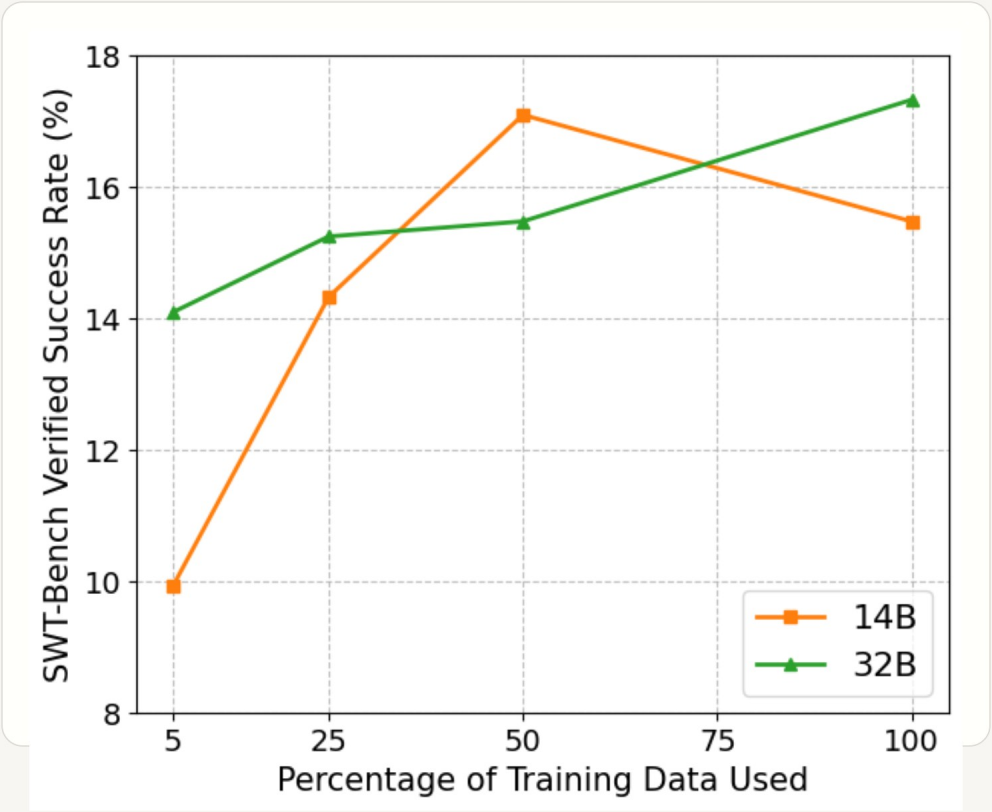
Fine-tuned model is capable of inference-time scaling

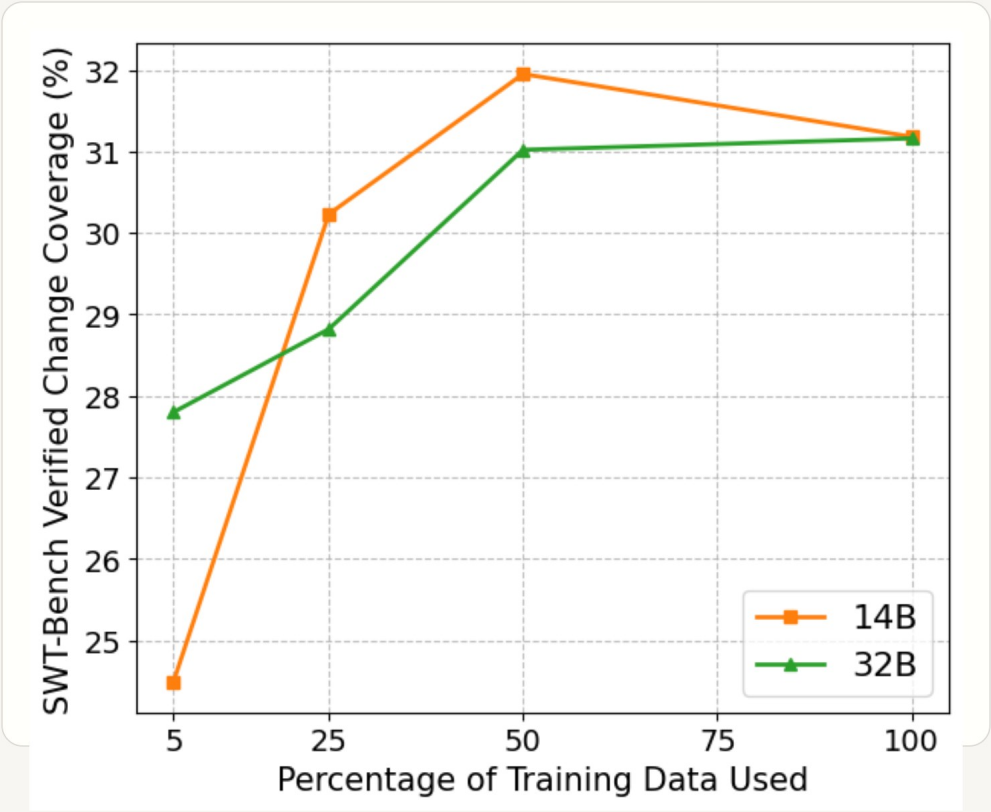




DATA SCALING

Bigger model keeps learning from data, while smaller model possibly overfits





LIMITATIONS AND NEXT STEPS

Where SWE-Tester stops today

Limitations

- Python only
- One test file per issue
- Heuristic data filters
- Training tests not execution-verified
- Test localization not trained

Next steps

- RL with execution feedback (F→P as reward)
- Stronger reasoning at inference
- Rerank by change coverage
- More languages and multi-file tests

SUMMARY

Open models can learn to write issue reproduction tests from developer-written PR tests alone

41K training instances · up to +10.4 pts success · +21 pts change coverage · gains across 3 model families

Thank you

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